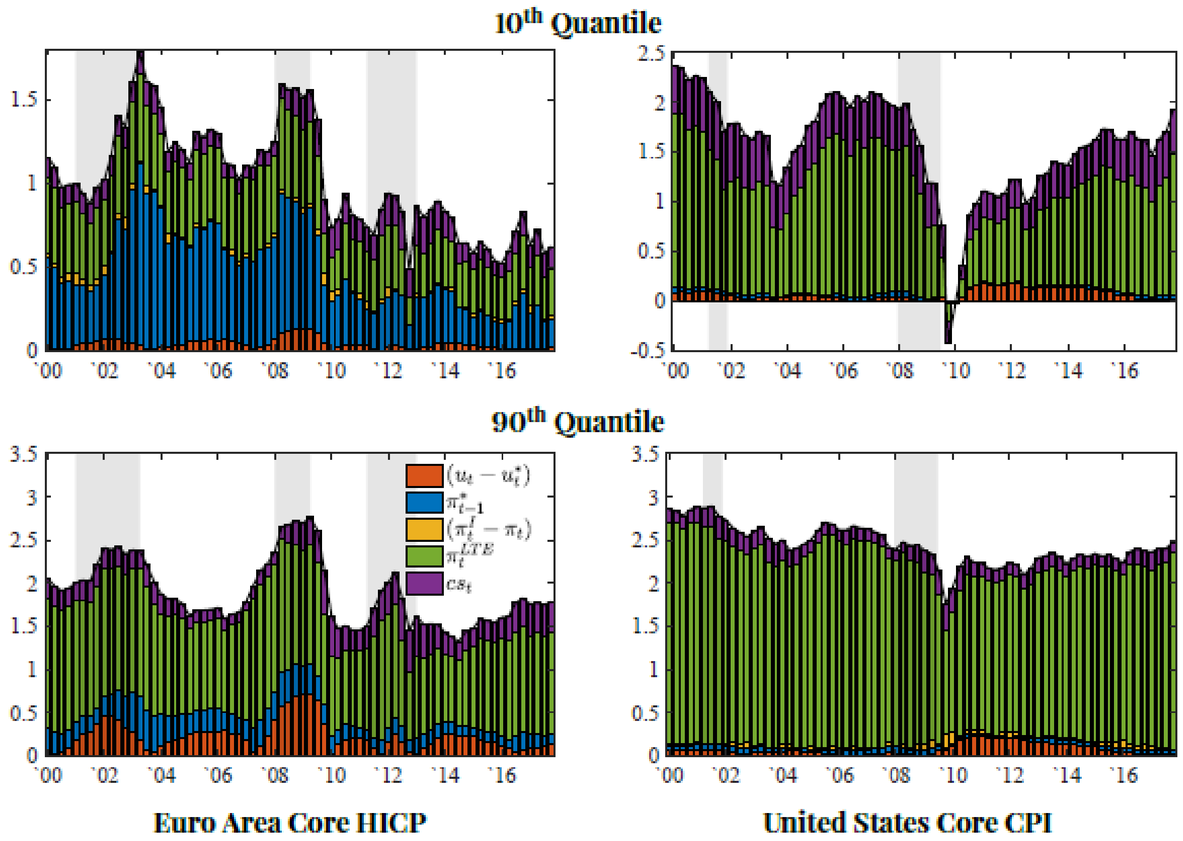
INFLATION AT RISK

WITH THE RECENT EPISODES OF LOW INFLATION ‘INFLATION-AT-RISK’ IS A KEY INDICATOR TO TRACK AND STUDY

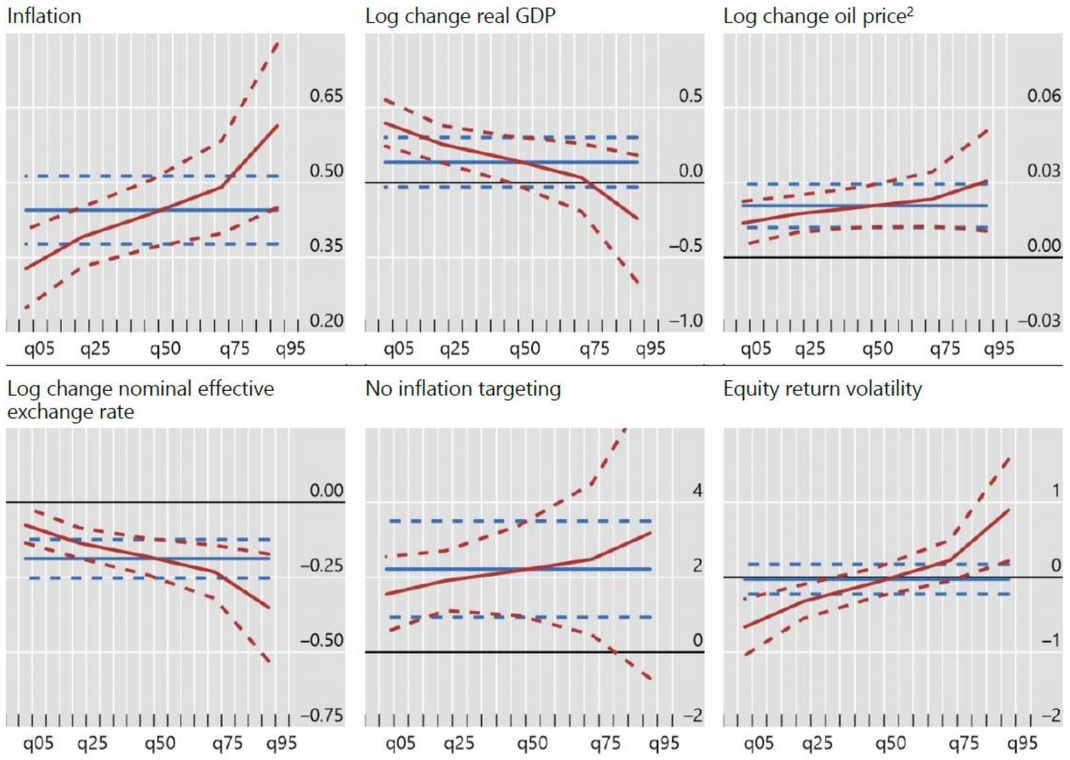
Figure 10: Historical Contributions of Economic and Financial Factors.



Source: Lopez-Salido and Loria (2020)

Estimated quantile regression and OLS coefficients, including financial conditions, EMEs

Graph 13



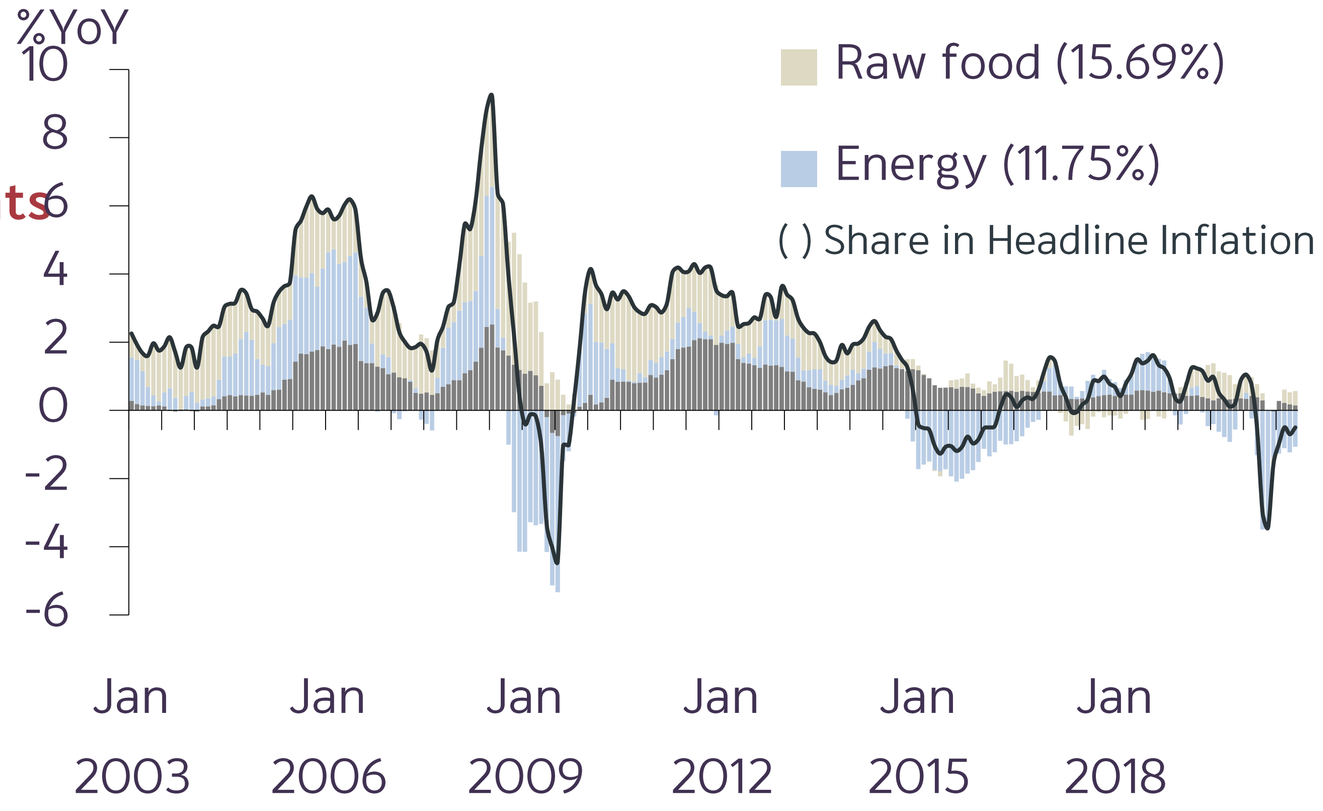
¹ OLS regressions in blue and quantile regressions in red; 90% confidence intervals. ² In domestic currency.

Source: Banerjee et.al. (2020)

DATA & ANALYSIS FRAMEWORK (1)

- **Monthly data (2003 onwards)**
- **Headline inflation with its components and 5 explanatory variable groups**
  1. Domestic economic activities
  2. Foreign factors
  3. Inflation expectations
  4. Financial conditions
  5. Component specific factors

Contribution of headline inflation



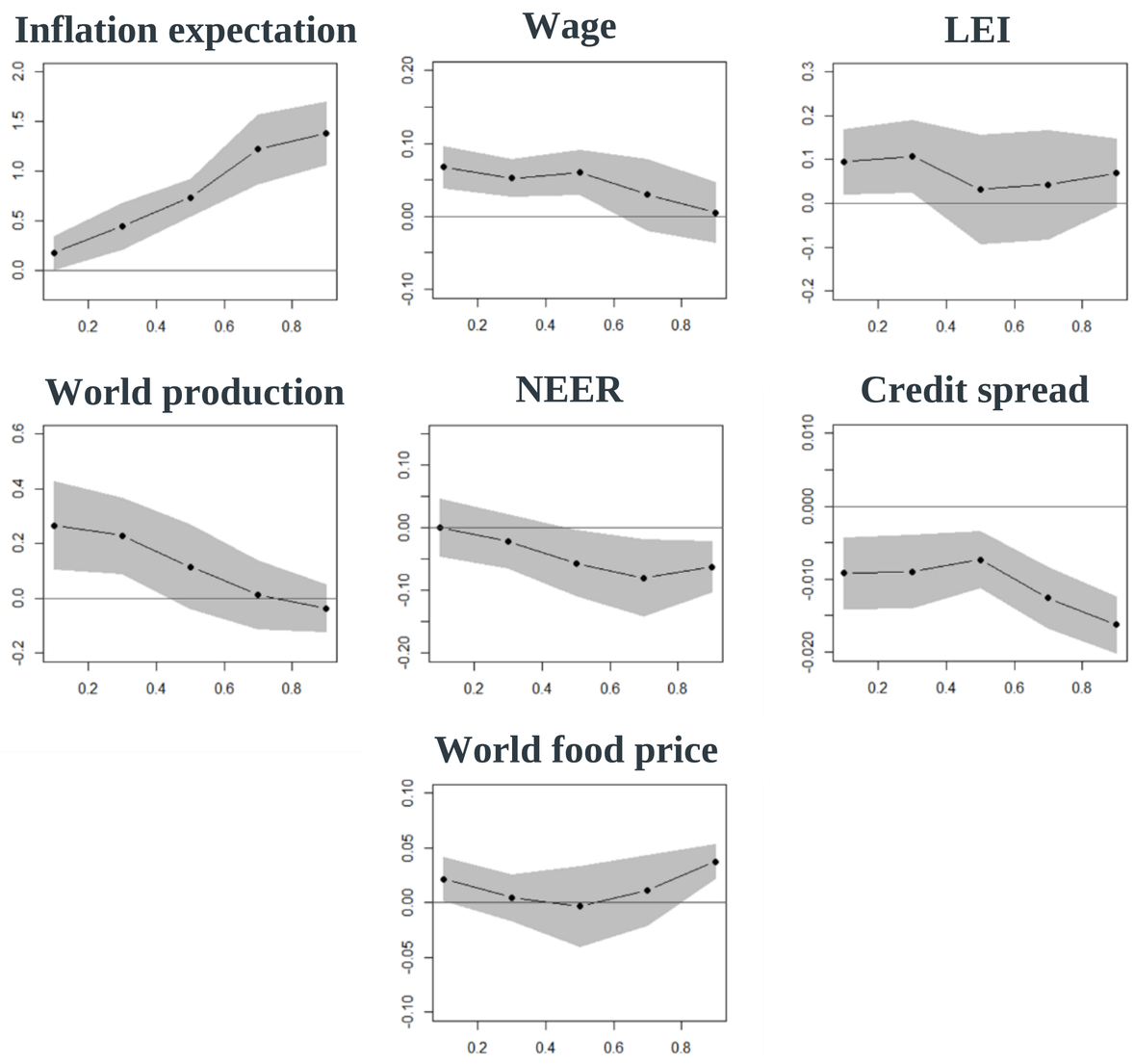
Source: Ministry of Commerce, calculated by Bank of Thailand

DATA & ANALYSIS FRAMEWORK (2)

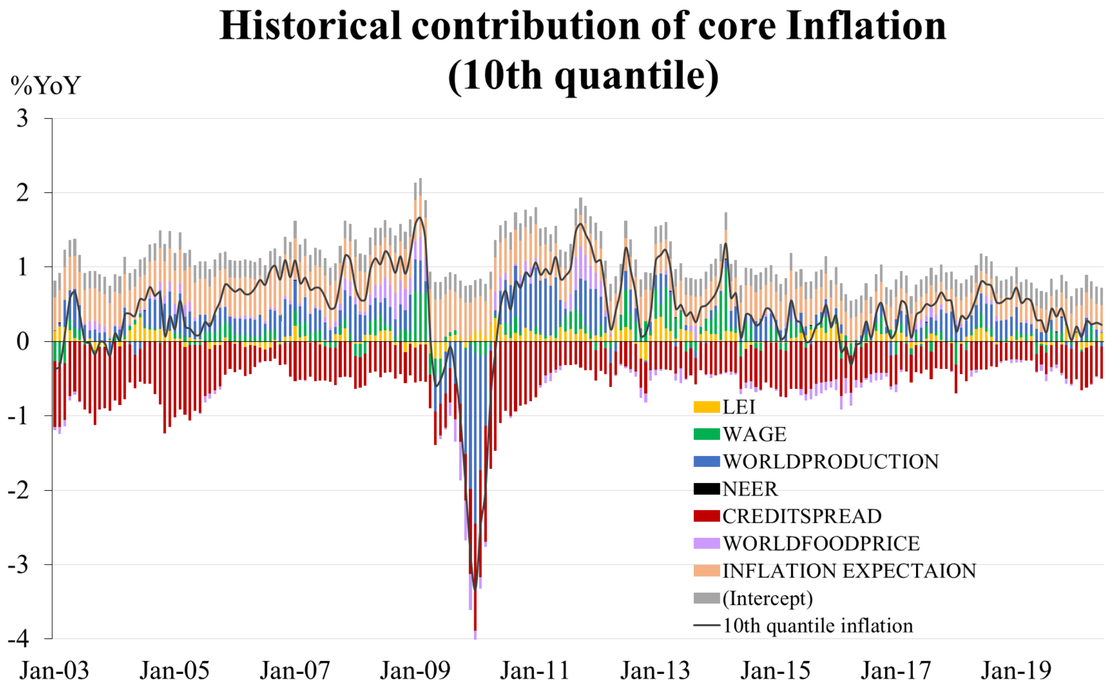
Variable selection	• Correlation table • Quantile LASSO / AIC stepwise selection • Judgement
Estimation	• Quantile regression
Robustness check	• Alternative variables • Alternative timeframes • Probability integral transform (PIT)
Output	• **Thailand inflation determinants (10th quantiles)** (core, raw food, energy and headline inflation) • **Historical inflation decomposition** • **Future conditional inflation distribution**

CORE INFLATION

Estimated quantile regression coefficients; 90% confidence interval

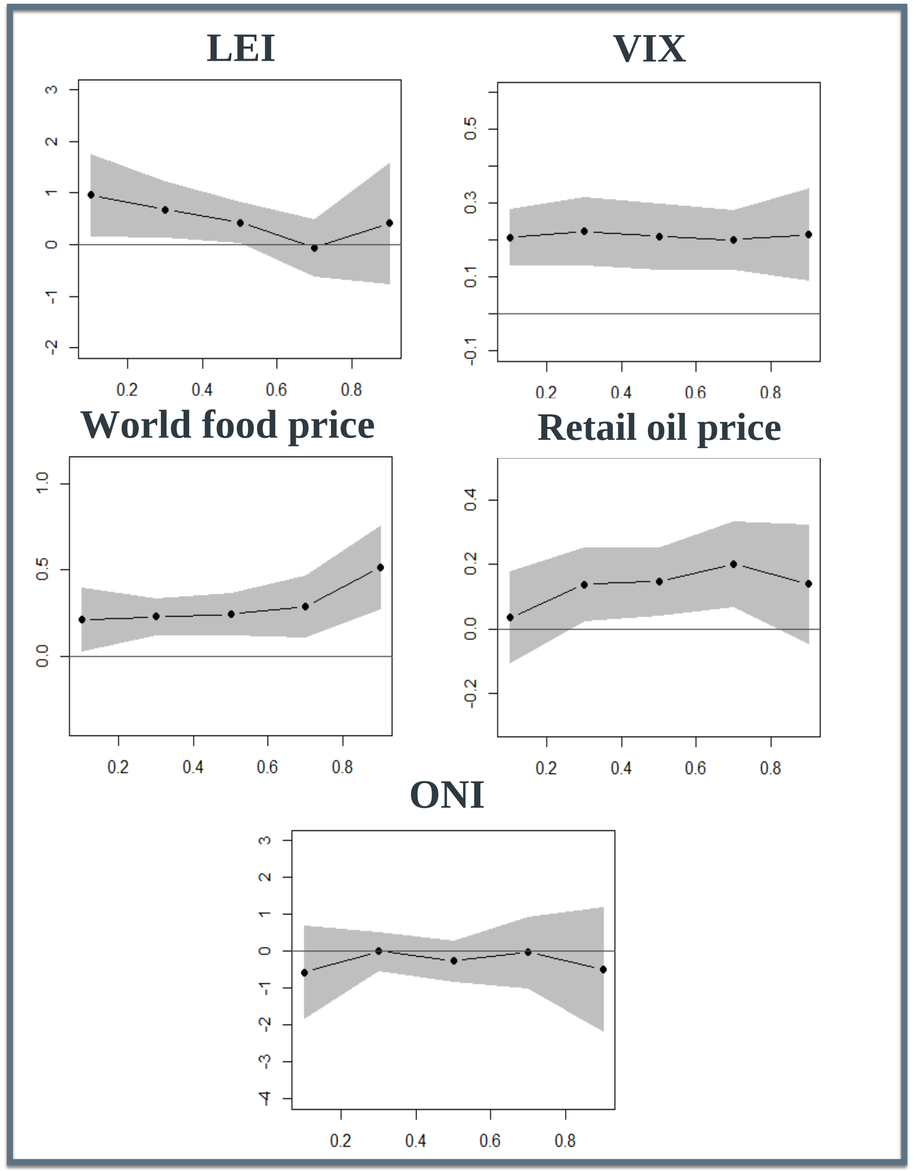


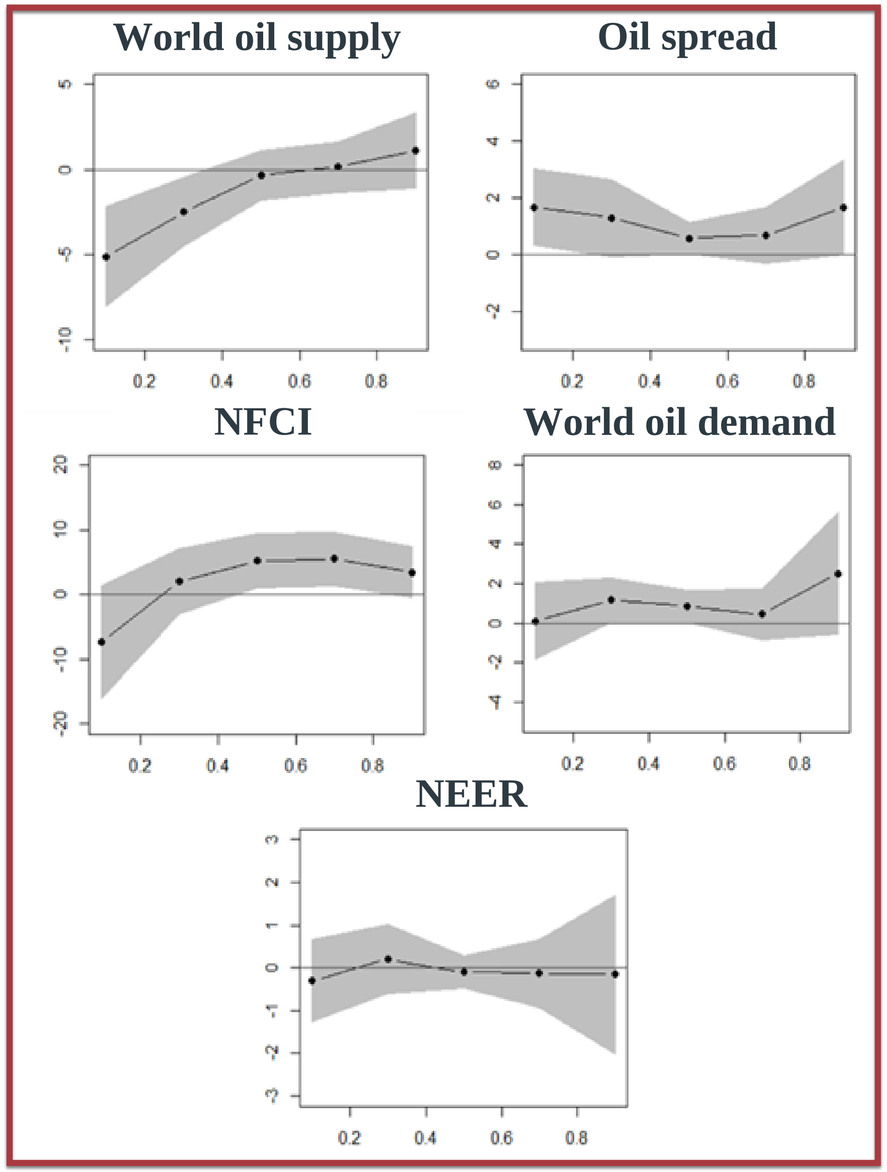
- **Inflation expectations'** anchoring effect is weaker when core inflation is low.
- However, the coefficient for **output** is significant and positive at the lower quantiles.



RAW FOOD PRICE & ENERGY PRICE INFLATION

Estimated quantile regression coefficients; 90% confidence interval

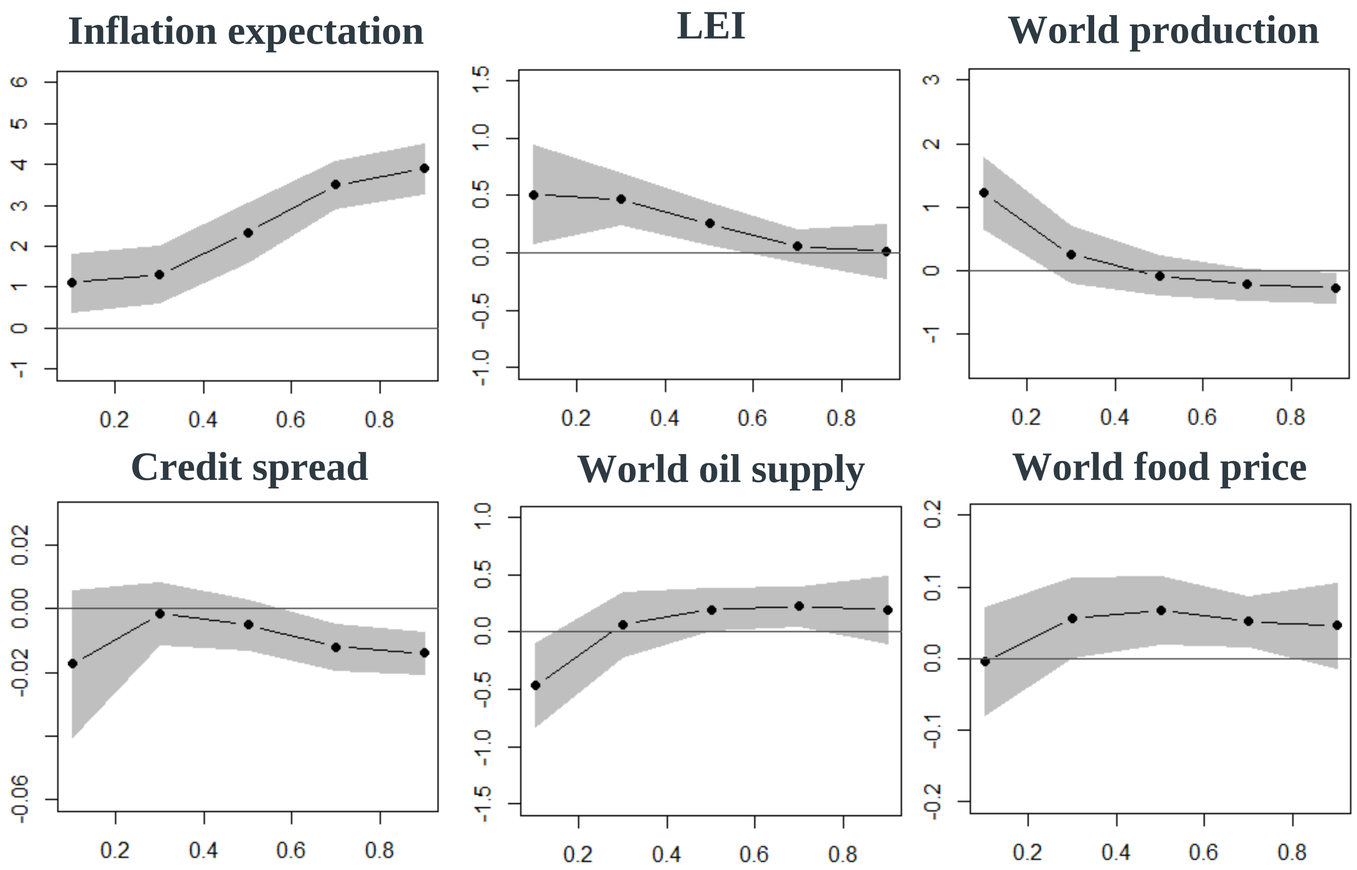




- **Raw food price at risk** is driven by world food prices and domestic output.
- **Increases in global oil supply and lower oil spreads** are drags on energy price inflation at risk.

HEADLINE INFLATION DETERMINANTS

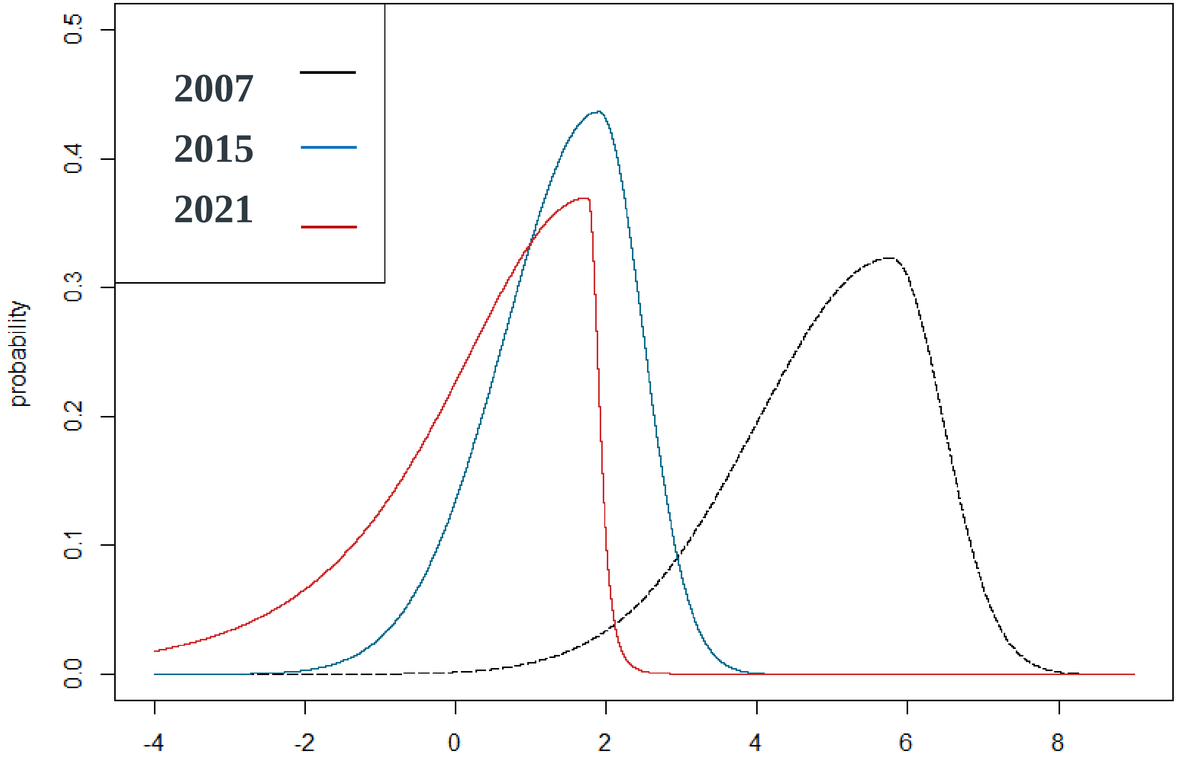
Estimated quantile regression coefficients; 90% confidence interval

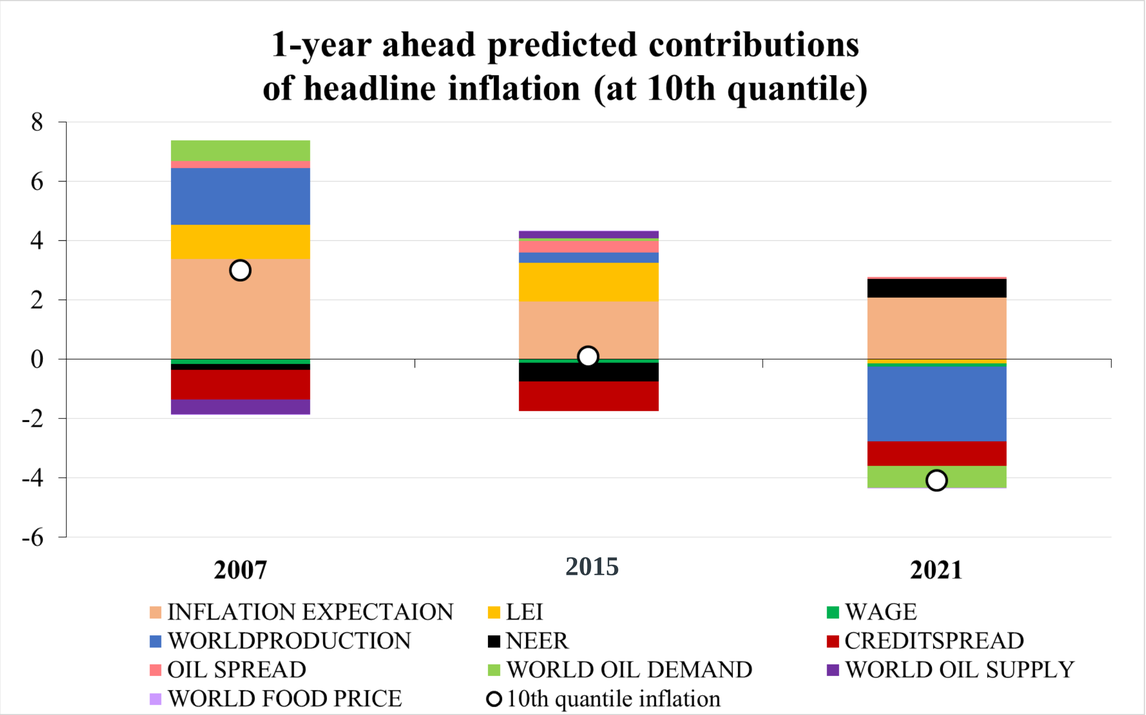


- Driven by results from core inflation, **inflation expectations** play a bigger role in the higher quantiles.
- **The Phillips curve relationship** (domestic and global) remains strong in the lower quantiles.

CONDITIONAL PREDICTIVE INFLATION DENSITIES

- The recent decline in global production is the biggest contributor to Thailand's inflation at risk in 2021.



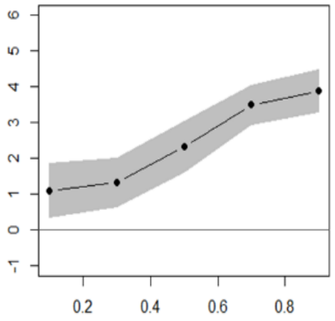


ROBUSTNESS CHECK: *ALTERNATIVE VARIABLES*

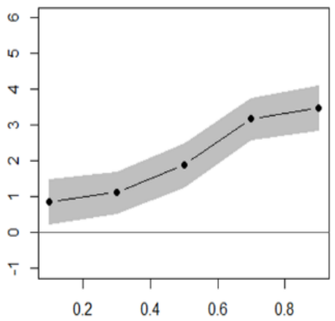
➢ The results are pretty robust to changes in the regression specifications.

Inflation expectation

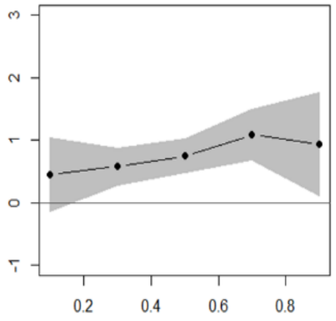
Inflation expectation 10Y



Inflation expectation 5Y

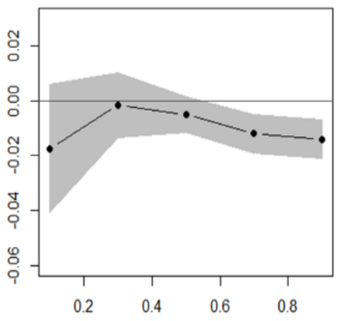


Inflation expectation 1Y concensus

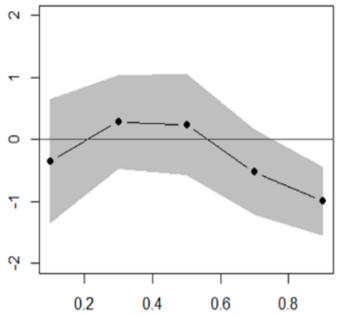


Financial conditions

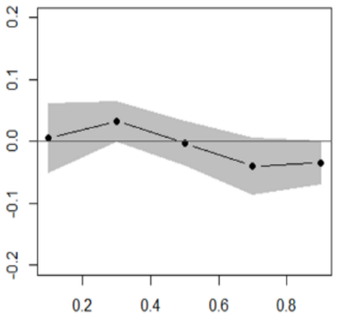
Credit spread



NFCI

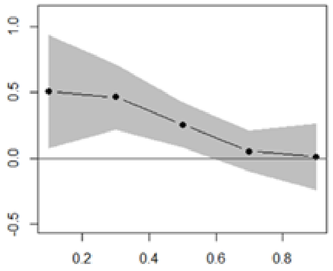


VIX

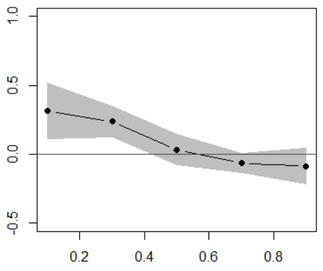


Domestic economic

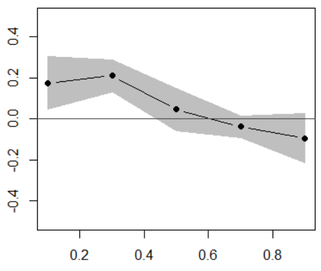
LEI



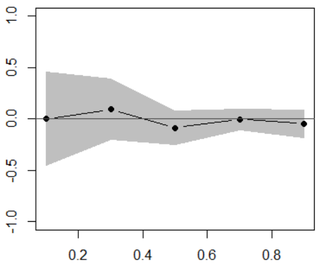
BSI expectation



BSI

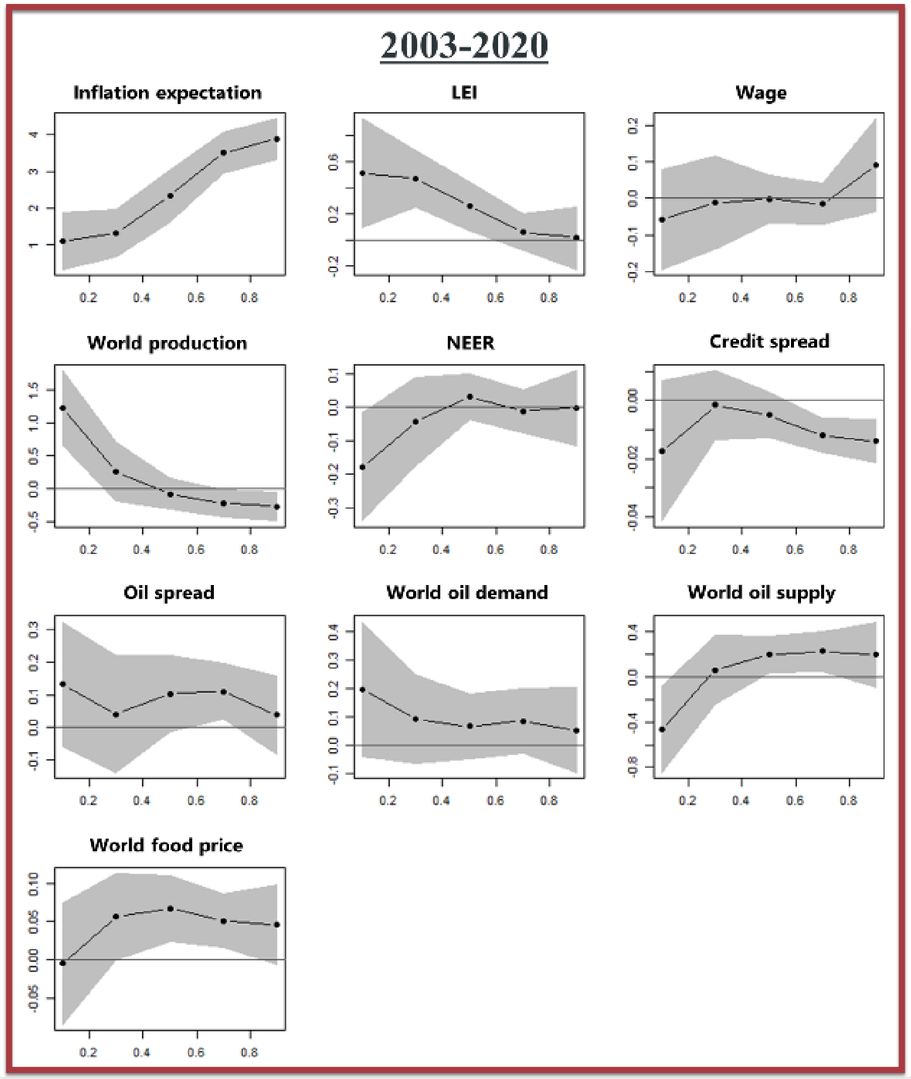


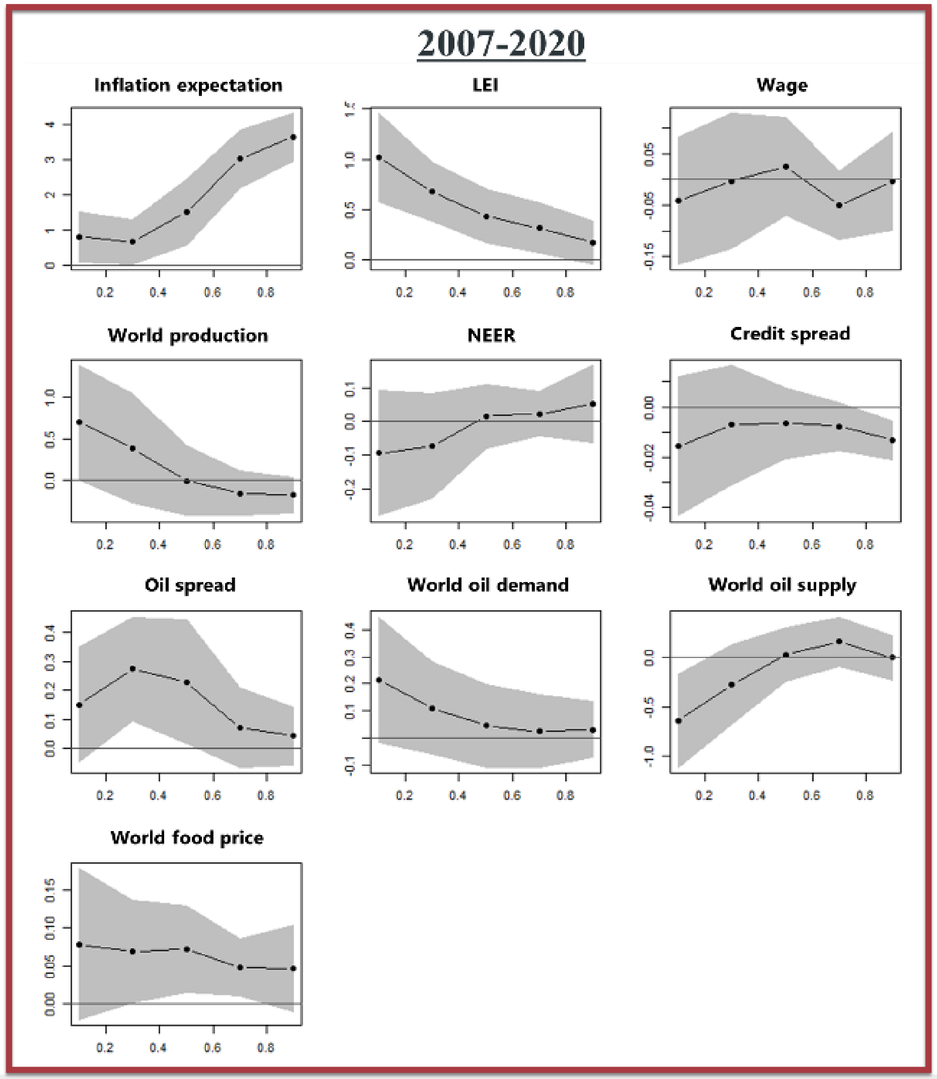
CEI



ROBUSTNESS CHECK: *ALTERNATIVE TIMEFRAMES*

- Results from earlier periods are driving the overall trends observed earlier, as there seem to be a structural break during the period of lower inflation since 2015.





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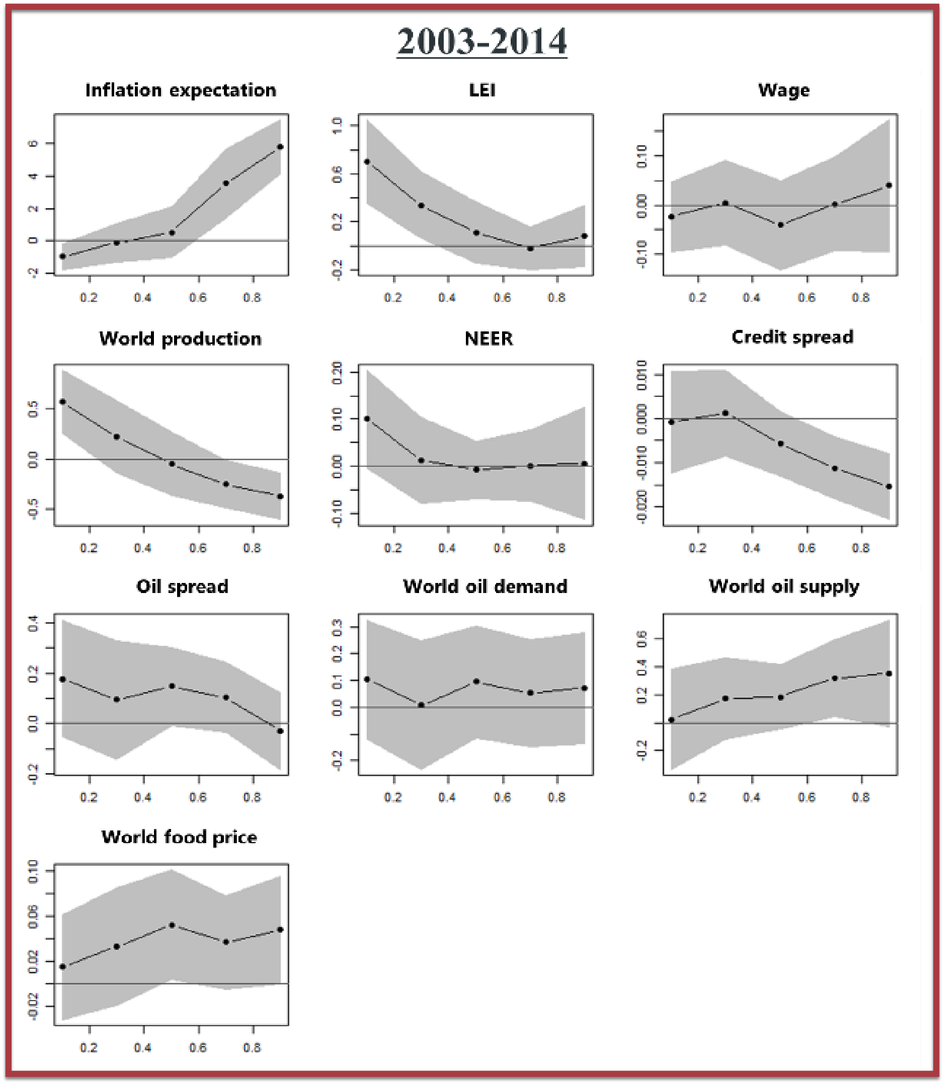
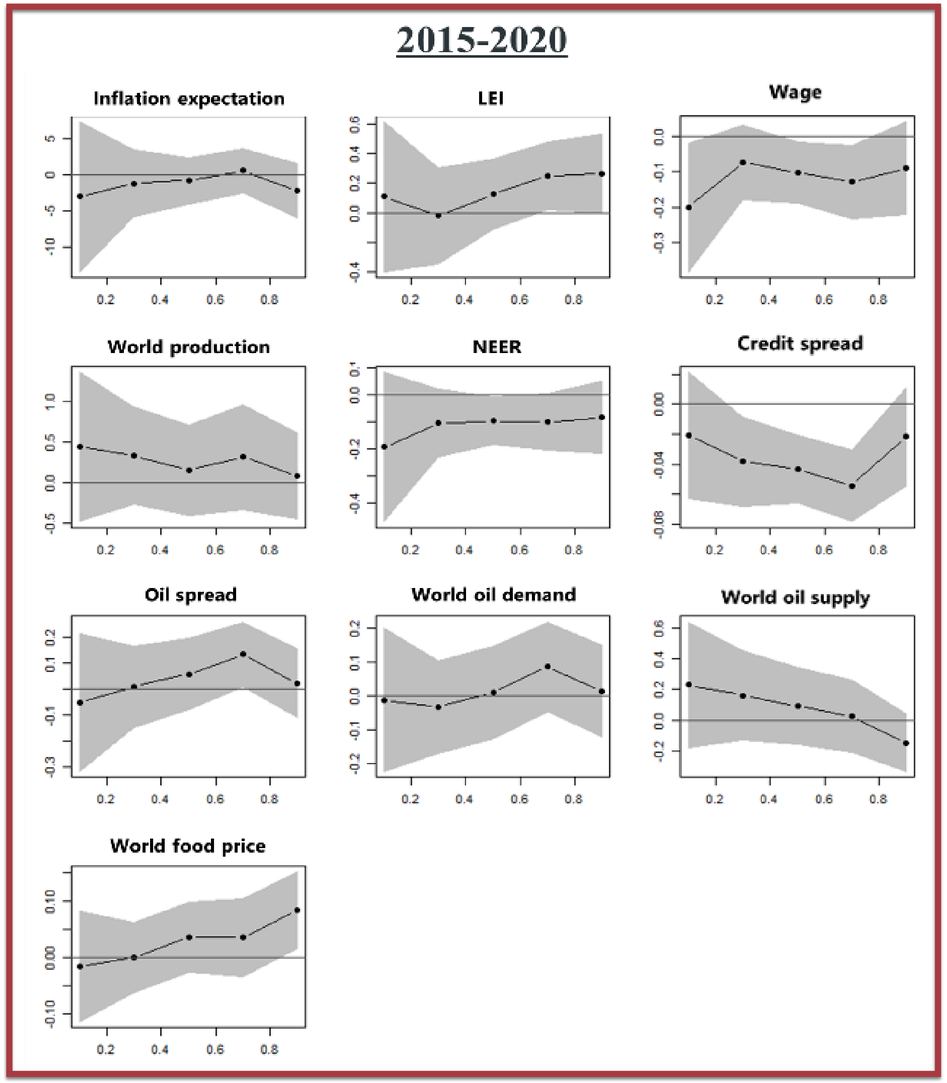
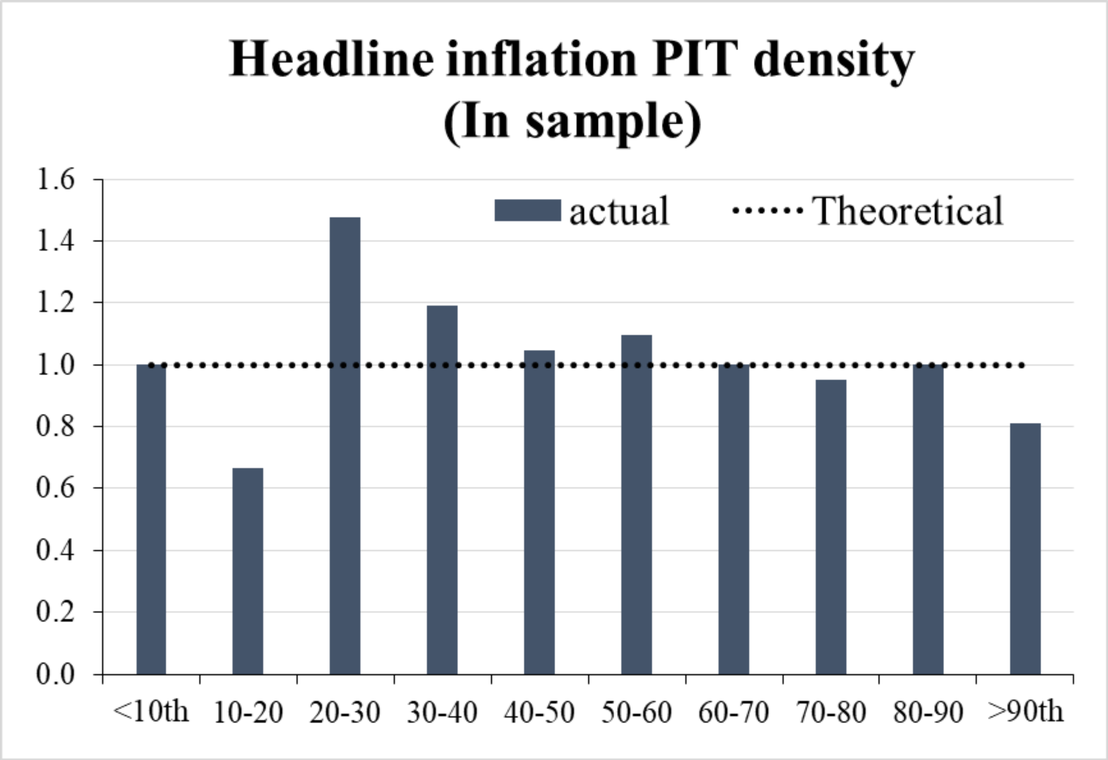
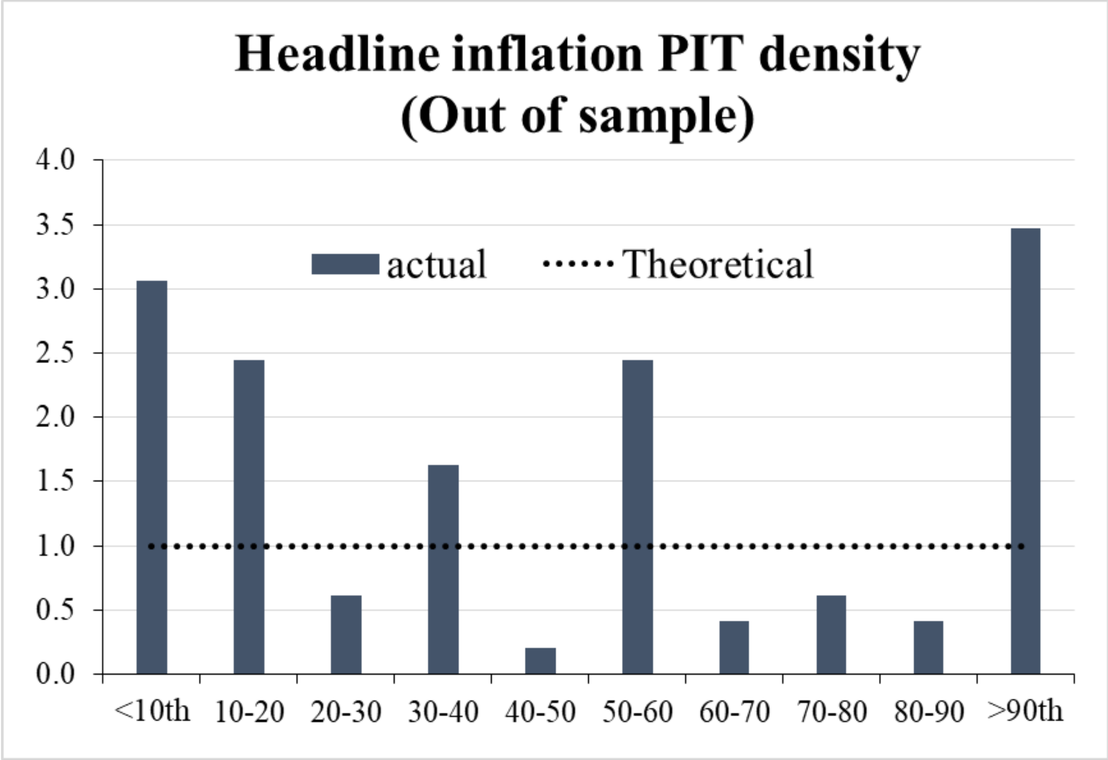




ROBUSTNESS CHECK: *PAST FORECASTING ACCURACY*

- We find that the forecasting accuracy is reasonable in-sample, but is quite unreliable out-of-sample.





CONCLUSIONS

1. **Inflation expectations** play an important role in anchoring future inflation, but that role becomes less important in the lower quantiles.

2. **Tightening financial conditions** are a drag to future inflation in all quantiles.

3. **The Phillips curve relationship** is strong in the lower quantiles.

4. To stimulate inflation in lower quantiles, **policy makers should focus on boosting economic activity**.

5. With the change in inflation environment in 2015 and the ongoing COVID pandemic, forecasting into the future should be done with extreme care.